

Soumyadeep Nath

Aspiring Data Scientist & AI Builder | Python · SQL · ML · GenAI · Full-Stack AI

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PROFESSIONAL SUMMARY

From literature to data — building the bridge deliberately. Currently completing an Executive PG Programme in Data Science & AI at IIT Roorkee, working hands-on with Python, SQL, Machine Learning, and Generative AI systems including RAG pipelines, embeddings, and LLM APIs. Built and deployed two live AI applications: **AlignResume**, a production-grade full-stack AI platform with truthfulness guardrails and structured JSON prompt pipelines, and **Future Fit**, a Streamlit-based skill trend analytics platform built on real job market data. A background in academic writing and language analysis provides a unique edge: translating complex model outputs into clear, compelling narratives that stakeholders understand.

EDUCATION & TRAINING

Executive PG Programme in Data Science & AI – In Progress

Feb 2026 – Present

Indian Institute of Technology (IIT) Roorkee | Remote

Coursework: Python · SQL · NumPy · Pandas · Machine Learning · Statistics · Data Visualization · Generative AI · Prompt Engineering · RAG · LLM APIs · Embeddings · Vector Databases

- Engineered data pipelines and automated ETL workflows using Python (Pandas, NumPy) to clean, transform, and prepare structured and unstructured datasets for downstream analysis.
- Queried relational databases using advanced SQL — window functions, joins, subqueries, and aggregations — extracting actionable, analysis-ready business insights.
- Constructed EDA frameworks applying statistical reasoning, hypothesis testing, and A/B testing methodology to validate data-driven decisions.
- Developed and evaluated supervised Machine Learning models — regression, classification, and tree-based ensembles (Random Forest, Gradient Boosting) — using scikit-learn.
- Implemented Generative AI applications by integrating LLM APIs, engineering structured prompts, and building RAG systems with embeddings and FAISS/Chroma vector databases.
- Translated complex model outputs into structured data narratives and interactive dashboards using Matplotlib, Seaborn, and Plotly; documented findings in reproducible Jupyter notebooks.

M.A. in English Literature — First Division

Oct 2022 – Jul 2024

Presidency University, Kolkata

B.A. Honours in English Literature — First Division

Sep 2019 – Jul 2022

Presidency University, Kolkata

PROJECTS

AlignResume — AI-Powered Resume Optimization Platform

May 2026

Live: align-resume-beta.vercel.app · GitHub: github.com/sdn9300/align-resume · Stack: Next.js · TypeScript · Tailwind CSS · Groq API (LLaMA 3.3 70B) · Playwright · Zod · Vercel

- Architected and deployed an end-to-end full-stack AI web application that parses resumes and job descriptions, scores alignment quality, and generates ATS-optimized bullet rewrites via a structured JSON prompt pipeline over the Groq API — with Zod schema validation ensuring deterministic, parseable LLM outputs.
- Implemented an automated truthfulness guardrail layer that validates rewritten bullets against the original resume, blocking export if hallucinated metrics, employers, or skills are detected — prioritizing honesty over inflation.
- Engineered Playwright-based headless PDF export, rate-limited API proxy (10 runs/hr per IP), and GitHub-Vercel CI/CD auto-deploy; deepened expertise in LLM application architecture, prompt engineering, and production-grade full-stack AI deployment.

Future Fit — AI-Powered Skill Trend Intelligence Platform

Jun 2026

Live: skilltrendanalysis-rf3zefgsjaa4l9f8pu2prb.streamlit.app · GitHub: github.com/sdn9300/Future-Fit-AI-Powered-Skill-Trend-Analysis

Stack: Python · Pandas · Plotly · Streamlit · Prophet · mlxtend (Apriori) · SciPy · Groq API (LLaMA 3.3 70B)

- Built an end-to-end Time Series and NLP analysis platform processing 224,605 skill mentions extracted from 50,000+ global AI/Data Science job postings (2020–2026), using a taxonomy-driven regex extraction engine with word-boundary matching

across 100+ skills.

- Engineered a 10-step deterministic cleaning pipeline and developed a Market Basket Analysis (Apriori) engine mapping 230 frequent skill itemsets and extracting association rules (lift ≥ 0.90 , confidence ≥ 0.38) to identify strong co-occurrence trends like PyTorch $\rightarrow$ TensorFlow.
- Implemented a tiered yearly forecasting model using Prophet to project demand shares into 2027, integrating a dynamic fallback to linear trend extrapolation if the 80% confidence interval width exceeded a $1.5\times$ variance threshold.
- Deployed a 6-panel Streamlit dashboard featuring Plotly charts, interactive association rules, and a Groq LLM-powered Skill Gap Advisor enriched with association rule insights to generate contextual learning paths.

Gleaner — Multi-Board Job Scraper & ETL Pipeline

Jun 2026

GitHub: github.com/sdn9300/gleaner-job-scout

Stack: Python · BeautifulSoup4 · Firecrawl API · Google Sheets API · YAML · Git

- Architected and implemented a modular ETL pipeline using the Adapter Design Pattern to concurrently fetch, normalize, and aggregate job listings from 4+ major platforms (Naukri, Indeed, RemoteOK, Wellfound/WWR).
- Engineered robust scraping mechanics using stealth configuration headers, custom bot user-agents, and Firecrawl API's structured extraction logic to ensure reliable, anti-bot-resistant data retrieval.
- Designed data processing stages utilizing exact and similarity-based deduplication algorithms, role-relevance keyword analysis, and location-based filters to clean raw unstructured payloads.
- Integrated automated reporting channels by establishing programmatic write pathways to local CSV outputs and secure synchronization with cloud-hosted spreadsheets using the Google Sheets API (gsread / google-auth).
- Developed a configurable CLI engine driven by centralized YAML files, allowing granular execution control over logging levels, output size limits, and targeted source execution.

SKILLS

Languages	Python, SQL, TypeScript
Frameworks & Tools	Next.js, Tailwind CSS, Streamlit, Vercel, Git / GitHub, BeautifulSoup4, Firecrawl API, YAML
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, SciPy, Prophet, mlxtend, Zod, gsread
Machine Learning	Regression, Classification, Random Forest, Gradient Boosting, Feature Engineering, Model Evaluation, Time Series Forecasting (Prophet), Association Rule Mining (Apriori / mlxtend)
Generative AI	LLM APIs (Groq, OpenAI), Prompt Engineering, Structured JSON Pipelines, RAG, Embeddings, Vector Databases (FAISS / Chroma), Guardrail Design
Data Analytics	EDA, Data Cleaning, ETL Pipelines, Web Scraping, Statistical Analysis, Hypothesis Testing, A/B Testing, Data Storytelling
Visualization	Matplotlib, Seaborn, Plotly, Power BI, Jupyter Notebook, Google Colab
Soft Skills	Critical Thinking, Analytical Reasoning, Communication, Attention to Detail, Adaptability

CERTIFICATIONS

Machine Learning Statistical Foundations Professional Certificate

LinkedIn Learning × Wolfram Research | Completed: Mar 2026 | Skills: Machine Learning · Linear Algebra · Wolfram Language

Anaconda Python for Data Science Professional Certificate

LinkedIn Learning × Anaconda | Completed: Mar 2026 | Skills: Python · Data Analysis · Statistics

Career Essentials in Generative AI

LinkedIn Learning × Microsoft | Completed: Mar 2026 | Skills: Generative AI · Microsoft Copilot · Responsible AI